

Homework 3

For this homework you will work with your group on a **GitHub** repo that creates a nice website to answer the questions below. One group member ('C') needs to turn in the link to this website!

You can use the same repo as HW 2 or start a new one. That is up to your group.

Assignment Tasks

Each group member should have a draft of their homework solution added to the repo by Monday evenings (the individual due date). The final, combined, submission must then be done by Wednesday evening.

For this homework, member 'C' is responsible for leading this process, doing a general quality check on the final submission, and actually submitting the assignment link on Moodle.

However you end up combining the files into a final submission, create clear guidelines you all agree on as a group and communicate often (if something comes up, let people know).

In the final document, answers to the questions below along with a discussion of what you are doing (via markdown) and comments in the code should be included.

Task 1: Conceptual Questions

On the exam, you'll be asked to explain some topics. How about some practice?! Create a markdown list with the following questions:

1. If your working directory is `myfolder/homework/`, what *relative* path would you specify to get the file located at `myfolder/MyData.csv`?
2. What are the major benefits of using **R** projects?
3. What is **Git** and what is **GitHub**?
4. What are the two main differences between a `tibble` and a `data.frame`?

Task 2 Reading Delimited Data

Note: Use chaining where needed/possible!

The data sets we'll use for this part comes from the [UCI machine learning repository](https://archive.ics.uci.edu/ml/).

Glass data

The first data set is called `glass.data`. You'll need to open the raw data set to determine the type of delimiter. The data is available at: <https://www4.stat.ncsu.edu/~online/datasets/glass.data>.

- The description of the data (not super useful - feel free to skip!):

Vina conducted a comparison test of her rule-based system, BEAGLE, the nearest-neighbor algorithm, and discriminant analysis. BEAGLE is a product available through VRS Consulting, Inc.; 4676 Admiralty Way, Suite 206; Marina Del Ray, CA 90292 (213) 827-7890 and FAX: - 3189. In determining whether the glass was a type of 'float' glass or not, the following results were obtained (# incorrect answers): Type of Sample Beagle NN DA Windows that were float processed (87) 10 12 21 Windows that were not: (76) 19 16 22 The study of classification of types of glass was motivated by criminological investigation. At the scene of the crime, the glass left can be used as evidence...if it is correctly identified!

- The variables and their descriptions:

Variable	Description
Id	Number 1-214
RI	Refractive index
Na	Sodium (unit measurement: weight percent in corresponding oxide, as are attributes 4-10)
Mg	Magnesium
Al	Aluminum
Si	Silicon
K	Potassium
Ca	Calcium
Ba	Barium
Fe	Iron

With the last variable being **Type of Glass** with values of - 1 building_windows_float_processed, - 2 building_windows_non_float_processed, - 3 vehicle_windows_float_processed, - 4 vehicle_windows_non_float_processed (none in this database), - 5 containers, - 6 tableware, - 7 headlamps.

1. Read this data into R directly from the URL using functions from the **tidyverse**. Notice that the data doesn't include column names - add those (in a manner of your choosing). Print out the **tibble** (just call the object name).

Yeast data

The second data set is called **yeast.data**. You'll need to open the raw data set to determine the type of delimiter. The data is available at: <https://www4.stat.ncsu.edu/~online/datasets/yeast.data>.

- The description of the data (not super useful! feel free to ignore):

The references below describe a predecessor to this dataset and its development. They also give results (not cross-validated) for classification by a rule-based expert system with that version of the dataset. Reference: 'Expert Sytem for Predicting Protein Localization Sites in Gram-Negative Bacteria', Kenta Nakai & Minoru Kanehisa, *PROTEINS: Structure, Function, and Genetics* 11:95-110, 1991.

- The variables and their descriptions:

Variable	Description
seq_name	Accession number for the SWISS-PROT database
mccg	McGeoch's method for signal sequence recognition.

Variable	Description
gvh	von Heijne's method for signal sequence recognition.
alm	Score of the ALOM membrane spanning region prediction program.
mit	Score of discriminant analysis of the amino acid content of the N-terminal region (20 residues long) of mitochondrial and non-mitochondrial proteins.
erl	Presence of 'HDEL' substring (thought to act as a signal for retention in the endoplasmic reticulum lumen). Binary attribute.
pox	Peroxisomal targeting signal in the C-terminus.
vac	Score of discriminant analysis of the amino acid content of vacuolar and extracellular proteins.
nuc	Score of discriminant analysis of nuclear localization signals of nuclear and non-nuclear proteins.
class	Localization site

2. Read this data into R directly from the URL using functions from the **tidyverse**. Notice that the data doesn't include column names - add those (in a manner of your choosing). Print out the **tibble** (just call the object name).

Task 3: Combining Excel and Delimited Data

The data set we'll use for this part comes from the [UCI machine learning repository](#). There are two data sets that are 'related to red and white variants of the Portuguese "Vinho Verde" wine.' There are physicochemical variables and a quality score, as rated by experts.

Input variables (based on physicochemical tests):

- fixed acidity
- volatile acidity
- citric acid
- residual sugar
- chlorides
- free sulfur dioxide
- total sulfur dioxide
- density
- pH
- sulphates
- alcohol

Output variable (based on sensory data):

- quality (score between 0 and 10)

3. There is an excel version of the white wine data set available at <https://www4.stat.ncsu.edu/~online/datasets/white-wine.xlsx>.
 - Download this file
 - Place it in a folder you know (such as your working directory for your project, the folder your .qmd file lives in, a data folder you've created in your repo, etc.)
 - Import the data from the first sheet using the **readxl** package
 - Print out the tibble (just call the object name)
4. When you print the data set out to the console, you may notice that some of the variable names are surrounded by backticks. This is because they are non-standard (they include a space in them). We can rename them in a number of ways. We'll do it by reading in the variable names from the 2nd sheet of the same file.

- Read in the data from the 2nd sheet. This should return a data frame with one column containing alternative versions of the variable names.
 - Grab that column and overwrite the current column names (`colnames()`) of your white wine `tibble`. Note that `colnames(object)` allows for ‘left hand side assignment’. That is, you can call it on the left and store an object in it to overwrite it. Similar to `names(object) <- c(...)`
5. Lastly, add a column to this data set to indicate the wines are white. That is, add a column that has values of ‘white’ for every observation. You can do this via the `$` operator or, if you know it already, the `mutate()` function from `dplyr`.
 6. There is a semi-colon delimited version of the red wine data set available at <https://www4.stat.ncsu.edu/~online/datasets/red-wine.csv>.
 - Read this in using the `readr` package. Be careful that the columns are read in as the correct type!
 - You should replace the variable names as done above
 - You should append a column denoting the `type` as “red”
 7. Combine these two data sets into one data set. They both have the exact same columns so this is an easy append task!
 - Use the `dplyr::bind_rows()` function (see the help) to create one `tibble` containing all of the wine data.

Task 4: Writing a Function

8. Write a basic function (call it `getIQR`) that takes in a vector of numbers and returns the interquartile range (75th percentile - 25th percentile). You can use the `quantile` function. Remove the name on the returned vector (i.e. make sure the returned item doesn’t have a name like 75% or something like that). Allow your function to pass unnamed arguments to the `quantile` function.
9. Take the object from question 7 and apply the `getIQR()` function to the `fixed_acidity` and `citric_acid` columns (two function calls).
10. Write another function (call it `getCoefVar`) that returns the sample coefficient of variation, defined as the sample standard deviation divided by the absolute value of the sample mean. Allow the function to pass unnamed arguments to `sd` and `mean`.
11. Repeat question 9 using the `getCoefVar()` function.
12. Create a function (call it `getSummaries`) that takes in a vector and returns both the IQR and the coefficient of variation. Do not rewrite your above two functions, call them. When returning your values, give them appropriate names. (This would be called a **wrapper** function.)
13. Repeat question 2 with the `getSummaries()` function.
14. Write your own **infix** function that can be used in replace of the `cbind` function (feel free to call the `cbind` function in your function :) Check that the function works on the two data frames below (`df1` & `df2`) and then all three objects (`df1`, `df2`, `vec1`)!

```
df1 <- data.frame(unif = runif(10), norm = rnorm(10))
df2 <- data.frame(gamma = rgamma(10, shape = 1), beta = rbeta(10, shape1 = 5, shape2 = 1))
vec1 <- rnorm(10, mean = 10)
```

To Turn In

Member ‘C’ should submit the link to the nicely rendered site on Moodle!